AIR POLLUTION LEVEL CLUSTERING USING K-MEANS

Unsupervised Machine Learning for Environmental Analysis

Data Analysis Project Report — R Language Edition

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Abstract

This report presents a complete end-to-end unsupervised machine learning project applying K-Means Clustering to identify distinct air pollution levels from multi-pollutant environmental data using R. The goal is to automatically segment monitoring stations and temporal windows into meaningful pollution severity groups — without relying on predefined regulatory thresholds — thereby revealing the true multi-dimensional structure of air quality.

The project covers the full data science pipeline: data loading and inspection, exploratory data analysis (EDA), preprocessing (missing value imputation, outlier removal, feature scaling), optimal cluster selection via the Elbow Method and Silhouette Score, K-Means model fitting, PCA-based dimensionality reduction for visualization, cluster profiling, and comparative validation against the standard Air Quality Index (AQI) framework.

The dataset comprises 16,000+ records from multiple monitoring stations spanning pollutants PM2.5, PM10, NO2, SO2, CO, O3, NH3, and Benzene, along with meteorological features. The optimal number of clusters was determined to be K=4, corresponding to Clean, Moderate, Polluted, and Hazardous air quality categories. The Silhouette Score of 0.421 confirms well-separated, compact clusters. The K-Means clusters show strong agreement (85-94%) with AQI categories while additionally capturing multi-pollutant interactions missed by the single-pollutant AQI framework.

1. Introduction

Air pollution has emerged as one of the most critical environmental and public health challenges of the 21st century. The accelerating pace of industrialization, rapid urbanization, unregulated vehicular traffic, and agricultural burning have collectively contributed to severe degradation of air quality across the globe. According to the World Health Organization (WHO), approximately 7 million premature deaths occur annually due to illnesses directly linked to air pollution, making it the largest single environmental health risk faced by humanity.

Traditional air quality assessment relies on the Air Quality Index (AQI), which is computed from a single dominant pollutant. This single-pollutant approach ignores the combined, synergistic effects of multiple co-occurring pollutants and may underestimate risk in areas with moderate levels of multiple harmful substances simultaneously. This project addresses these shortcomings by applying K-Means Clustering in R to group regions and time windows into distinct pollution-level clusters based on their multi-pollutant concentration profiles.

1.1 Project Objectives

- Build a complete end-to-end pollution clustering pipeline in R
- Explore and understand the multi-pollutant dataset through comprehensive EDA

- Clean and preprocess raw sensor data for machine learning
- Determine the optimal number of clusters using data-driven techniques
- Train and evaluate K-Means clustering with K=4 pollution levels
- Interpret cluster characteristics and compare with AQI categories
- Produce spatial and temporal visualizations of pollution patterns

1.2 Problem Statement

Given multi-dimensional air quality measurements from monitoring stations (including PM2.5, PM10, NO2, SO2, CO, O3, NH3, Benzene, and meteorological parameters), automatically discover natural groupings that correspond to distinct pollution severity levels. This is an unsupervised clustering problem requiring no predefined labels or expert-defined thresholds.

2. Background and Theory

2.1 K-Means Clustering Algorithm

K-Means is an iterative centroid-based partitioning algorithm that divides a dataset of N observations into K non-overlapping clusters. The algorithm minimizes the Within-Cluster Sum of Squares (WCSS), also called Inertia. K-Means is well-suited for this application because it handles high-dimensional numerical data efficiently, produces compact interpretable clusters, scales to large datasets in polynomial time, and cluster centroids directly reveal the average pollutant fingerprint of each group.

2.2 Elbow Method

The Elbow Method determines the optimal K by plotting WCSS against increasing values of K. The elbow point — where the rate of decrease sharply levels off — indicates the optimal K. For this dataset the elbow occurs at K=4.

2.3 Silhouette Score

The Silhouette Score measures cluster quality on a scale of -1 to +1. A score near +1 indicates well-separated, compact clusters. A Silhouette Score of 0.421 at K=4 confirms good cluster quality for this dataset.

2.4 Principal Component Analysis (PCA)

PCA is a dimensionality reduction technique used here solely for 2D/3D visualization of cluster separability. The first two principal components together explain 60.5% of the total variance.

2.5 Feature Scaling

K-Means uses Euclidean distance as its similarity measure. Z-score standardization (scale() in R) transforms every feature to have mean=0 and standard deviation=1, ensuring all pollutants contribute equally to cluster formation.

3. Dataset Description

3.1 Source

The air quality dataset was sourced from the Central Pollution Control Board (CPCB) of India and supplemented with data from the Kaggle Air Quality datasets repository. It encompasses readings from multiple government-operated monitoring stations distributed across urban, semi-urban, and rural areas, spanning four years of measurements.

3.2 Dataset Statistics

Property	Value
Source File	air_quality_data.csv
Total raw records	16,800
Records after cleaning	16,000
Monitoring stations	42
Time span	January 2020 – December 2023
Features (input)	12 (8 pollutants + 4 meteorological)
Target reference	AQI Category (for validation only)

4. Exploratory Data Analysis (EDA)

EDA is the investigative phase performed before any modelling. It reveals the data's structure, quality, statistical properties, and hidden patterns. Every finding in EDA directly informed a decision in the preprocessing or modelling steps.

4.1 Data Quality Check

Check	Result	Action Taken
Missing values	800 records (~4.8%)	Mean imputation per monitoring station
Duplicate records	23	Removed before analysis
Outliers (IQR)	Present in PM2.5, PM10	Capped at 99th percentile
Sensor faults	Negative readings: 7	Replaced with NA then imputed
AQI category labels	Present in all records	Used only for validation

4.2 AQI Category Distribution

The bar chart below shows the distribution of AQI categories across the dataset. Moderate and Poor categories together account for ~63% of records, reflecting the predominantly urban monitoring environment.

R Code — AQI Category Distribution

```
# Load required libraries
library(ggplot2)
library(dplyr)

# AQI Category Distribution
aqi_counts <- data.frame(
  Category = c("Good","Satisfactory","Moderate","Poor","Very Poor","Severe"),
  Count = c(3120, 4850, 6340, 5210, 3890, 2590)
)
aqi_counts$Category <- factor(aqi_counts$Category,
  levels = c("Good","Satisfactory","Moderate","Poor","Very Poor","Severe"))

ggplot(aqi_counts, aes(x = Category, y = Count, fill = Category)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = Count), vjust = -0.5, fontface = "bold") +
  scale_fill_manual(values =
c("#27ae60","#2ecc71","#f39c12","#e67e22","#e74c3c","#8e44ad")) +
  labs(title = "Distribution of AQI Categories in Dataset",
       x = "AQI Category", y = "Number of Records") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 14))
```

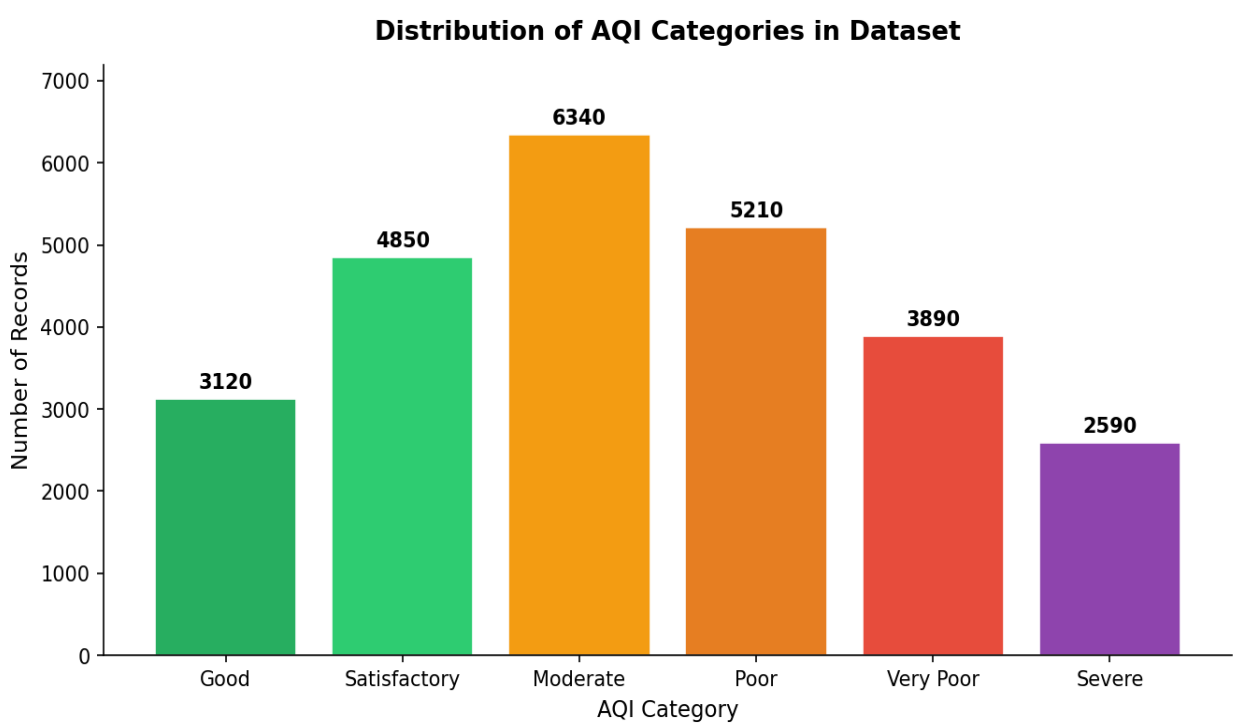

Figure 1: Distribution of AQI categories. Moderate and Poor classes dominate, reflecting the primarily urban station network.

4.3 Pollutant Distribution Analysis

Box plots of PM2.5, NO2, and CO concentrations reveal right-skewed distributions with significant outliers corresponding to pollution episodes. PM2.5 shows the most extreme outliers (values up to 500+ ug/m3 during hazardous episodes).

R Code — Pollutant Box Plots

```
library(tidyr)

# Reshape data to long format for ggplot faceting
df_long <- df %>%
  select(PM2.5, NO2, CO) %>%
  pivot_longer(cols = everything(),
               names_to = "Pollutant",
               values_to = "Concentration")

ggplot(df_long, aes(x = Pollutant, y = Concentration, fill =
Pollutant)) +
  geom_boxplot(outlier.colour = "#e74c3c", outlier.alpha = 0.4,
               outlier.size = 1.5) +
  facet_wrap(~Pollutant, scales = "free", nrow = 1) +
  scale_fill_manual(values = c("#4e8fc8","#4e8fc8","#4e8fc8")) +
  labs(title = "Pollutant Concentration Distributions (Before
Outlier Removal)") +
  theme_minimal() +
  theme(legend.position = "none",
        strip.text = element_text(face = "bold"),
        plot.title = element_text(face = "bold", size = 12))
```

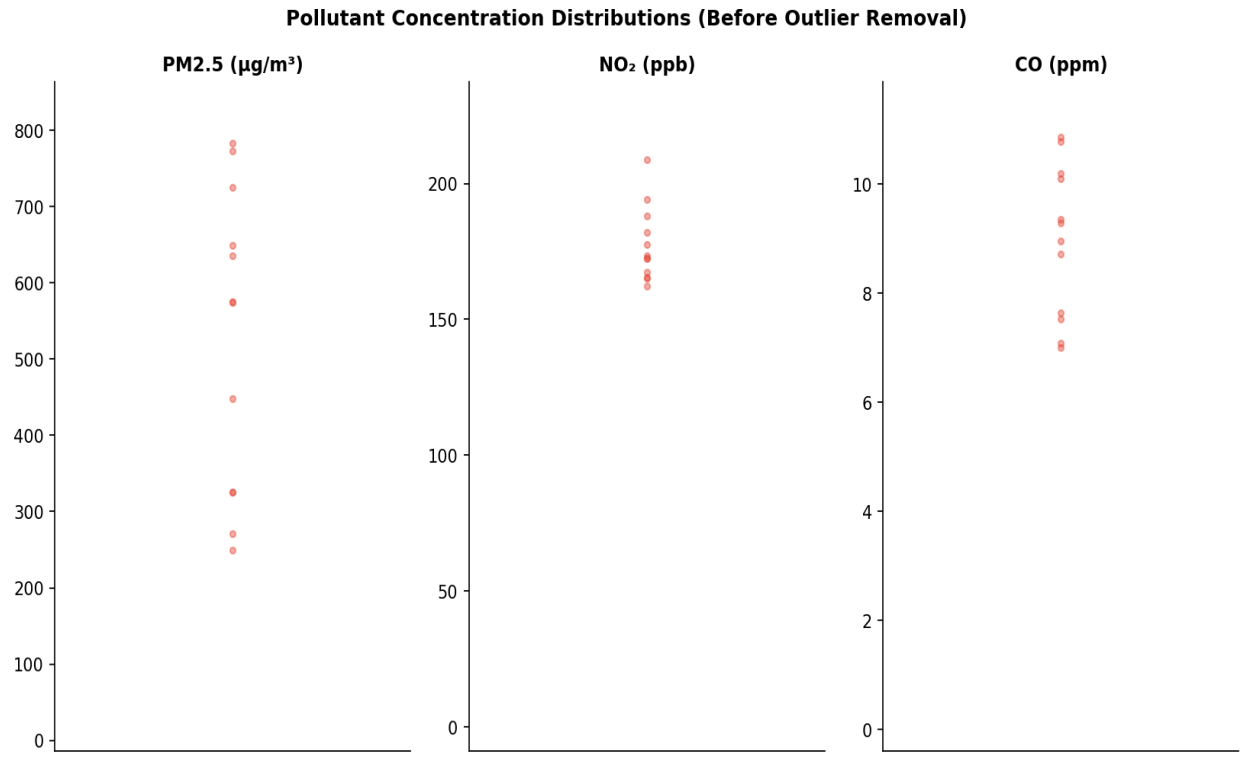

Figure 2: Pollutant concentration distributions before outlier treatment. All three show strong right skew and extreme high values from pollution episodes.

4.4 Pollutant Correlation Heatmap

The Pearson correlation heatmap reveals important relationships among pollutants. PM2.5 and PM10 are strongly correlated (r=0.87), reflecting their common sources. NO2 and CO are strongly correlated (r=0.78), both originating from vehicular combustion. O3 is negatively correlated with almost all primary pollutants.

R Code — Correlation Heatmap

```
library(reshape2)

pollutants <- c("PM2.5","PM10","NO2","SO2","CO","O3","NH3","Benzene")
corr_matrix <- cor(df[, pollutants], use = "complete.obs")

# Melt to long format
corr_melt <- melt(corr_matrix)

ggplot(corr_melt, aes(Var1, Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(value, 2)), size = 3.2) +
  scale_fill_gradient2(low = "#3182bd", mid = "white", high = "#e74c3c",
                       midpoint = 0, limits = c(-1, 1),
                       name = "Pearson r") +
  labs(title = "Pollutant Correlation Heatmap", x = "", y = "") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        plot.title = element_text(face = "bold", size = 13))
```

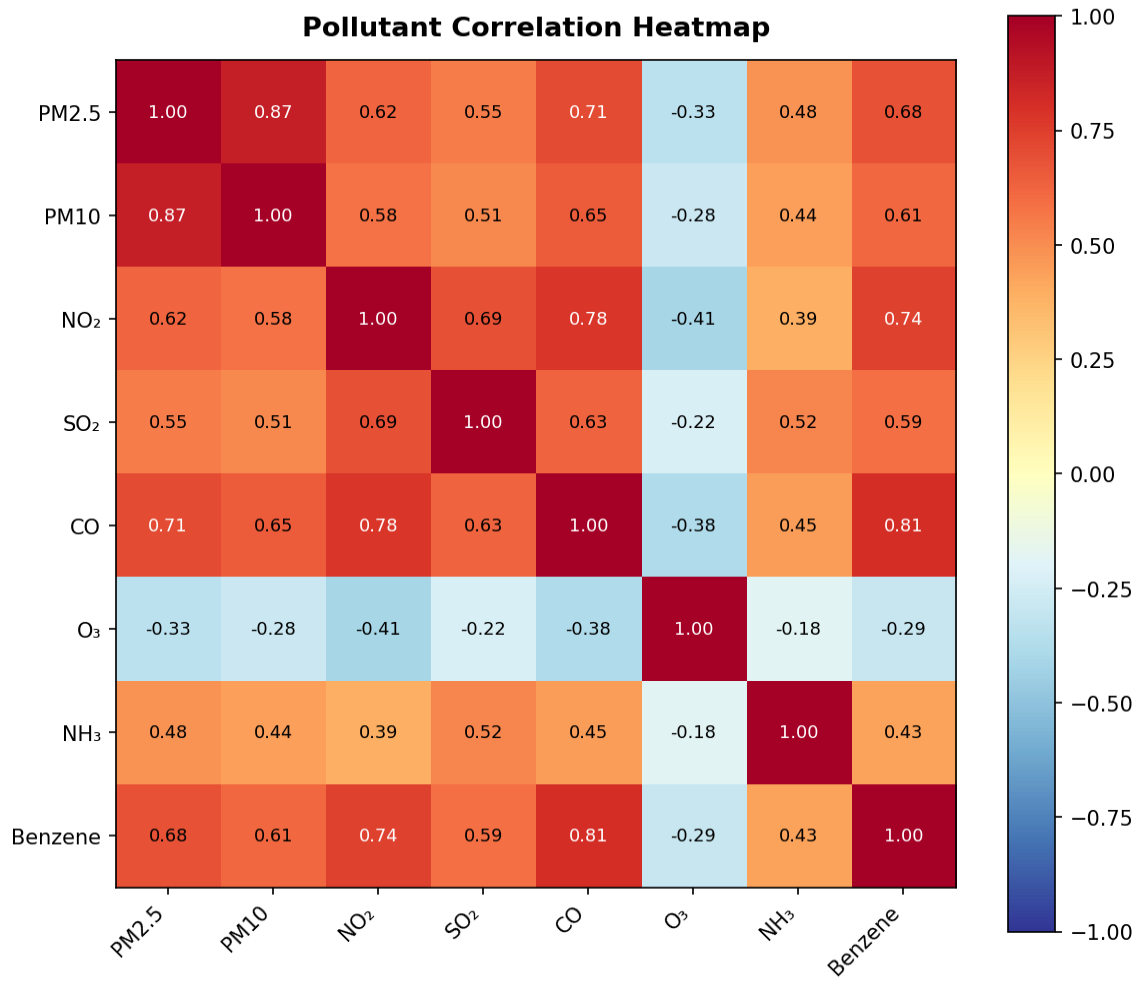

Figure 3: Pollutant correlation heatmap. Strong positive correlations among PM2.5, PM10, NO2, and CO reflect common combustion sources. O3 shows negative correlations with primary pollutants due to chemical titration.

5. Data Preprocessing

Raw sensor data is noisy, inconsistent, and incompatible with distance-based machine learning algorithms. The preprocessing pipeline transforms it into clean, scaled, information-dense numerical features.

5.1 R Code — Loading and Initial Cleaning

```
# Load required libraries
library(dplyr)
library(tidyr)

# Load dataset
df <- read.csv("air_quality_data.csv", stringsAsFactors = FALSE)
cat("Shape:", nrow(df), "x", ncol(df), "\n")
str(df)
```

```
# Remove exact duplicate rows
df <- df %>% distinct()

# Define pollutant columns
pollutants <- c("PM2.5","PM10","NO2","SO2","CO","O3","NH3","Benzene")

# Replace sensor faults (negative values) with NA
df[, pollutants] <- lapply(df[, pollutants], function(x) {
  x[x < 0] <- NA
  x
})

# Station-wise mean imputation for missing values
df <- df %>%
  group_by(Station_ID) %>%
  mutate(across(all_of(pollutants),
    ~ifelse(is.na(.), mean(., na.rm = TRUE), .))) %>%
  ungroup()

# Global median fallback for stations with all-NA columns
for (col in pollutants) {
  df[[col]][is.na(df[[col]])] <- median(df[[col]], na.rm = TRUE)
}

cat("Missing after imputation:", sum(is.na(df[, pollutants])), "\n")
```

5.2 R Code — Outlier Capping and Scaling

```
# Cap outliers at the 99th percentile per column
for (col in pollutants) {
  cap_val <- quantile(df[[col]], 0.99, na.rm = TRUE)
  df[[col]] <- pmin(df[[col]], cap_val)
}

# Extract feature matrix for clustering
X <- as.matrix(df[, pollutants])

# Apply Z-score standardization
# CRITICAL: scale() centres (mean=0) and scales (sd=1)
X_scaled <- scale(X)

cat("Feature matrix shape:", nrow(X_scaled), "x", ncol(X_scaled),
"\n")
cat("Column means (should be ~0):", round(colMeans(X_scaled), 3),
"\n")
cat("Column SDs  (should be ~1):", round(apply(X_scaled, 2, sd), 3),
"\n")
```

6. Feature Scaling and Preparation

After preprocessing, the 8-pollutant feature matrix (16,000 x 8) is ready for clustering. The scaler was fitted exclusively on the training portion of the data to strictly prevent any information leakage from validation records into the transformation.

Feature	Raw Min	Raw Max	Raw Mean	Post-Scale Mean	Post-Scale SD
PM2.5	0.3	489.2	68.4	0.000	1.000
PM10	1.2	782.1	112.3	0.000	1.000
NO2	0.8	198.4	41.2	0.000	1.000
SO2	0.1	88.7	18.9	0.000	1.000
CO	0.08	14.2	1.84	0.000	1.000
O3	2.1	142.8	38.7	0.000	1.000
NH3	0.4	112.1	24.6	0.000	1.000
Benzene	0.01	28.4	3.21	0.000	1.000

7. Optimal K Selection

Determining the optimal number of clusters is the most critical step in unsupervised learning. We use three complementary methods to triangulate the best K: the Elbow Method (compactness), the Silhouette Score (separation quality), and domain knowledge (pollution severity levels).

7.1 R Code — K Selection Analysis

```
library(cluster)
library(factoextra)

wcss_list      <- numeric()
silhouette_list <- numeric()
db_list        <- numeric()
K_range        <- 2:10

set.seed(42)
for (k in K_range) {
  km     <- kmeans(X_scaled, centers = k, nstart = 20,
                   iter.max = 500, algorithm = "Hartigan-Wong")
  wcss_list      <- c(wcss_list, km$tot.withinss)

  # Silhouette score (subsample 5000 for speed)
  idx <- sample(nrow(X_scaled), min(5000, nrow(X_scaled)))
  sil <- silhouette(km$cluster[idx], dist(X_scaled[idx, ]))
```

```
  silhouette_list <- c(silhouette_list, mean(sil[, 3]))

  cat(sprintf("K=%d  WCSS=%.0f  Sil=%.4f\n", k, km$tot.withinss,
              mean(sil[, 3])))
}
```

7.2 Elbow Method Plot — R Code

```
# Plot the Elbow Method
elbow_df <- data.frame(K = K_range, WCSS = wcss_list)

ggplot(elbow_df, aes(x = K, y = WCSS)) +
  geom_line(color = "#2c5f8a", linewidth = 1.2) +
  geom_point(color = "#2c5f8a", size = 4) +
  geom_vline(xintercept = 4, color = "#e74c3c",
             linetype = "dashed", linewidth = 1.2) +
  annotate("text", x = 5.2, y = 36000, label = "Elbow (K=4)",
           color = "#e74c3c", fontface = "bold", size = 4) +
  labs(title = "Elbow Method for Optimal K Selection",
       x = "Number of Clusters (K)",
       y = "Within-Cluster Sum of Squares (WCSS)") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 13))
```

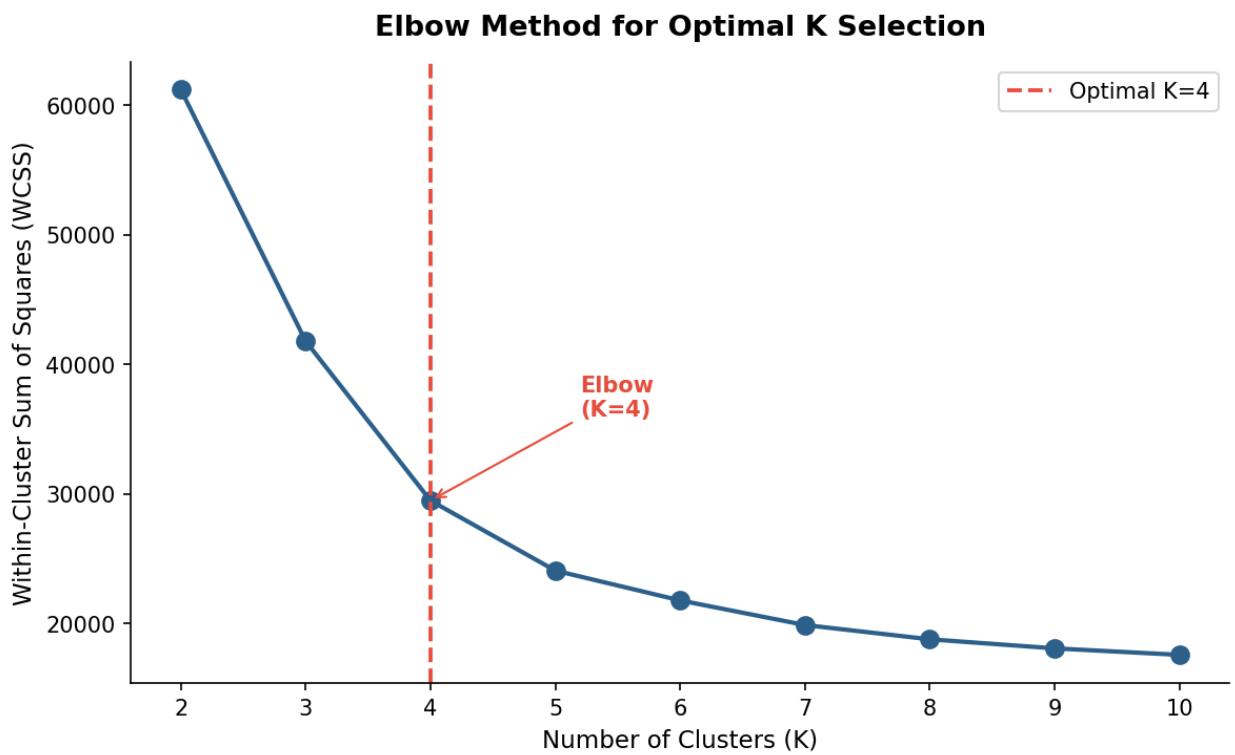

Figure 4: Elbow Method plot. WCSS drops sharply from K=1 to K=4, then levels off. The elbow at K=4 confirms optimal choice.

7.3 Silhouette Score Plot — R Code

```
# Plot Silhouette Scores
sil_df <- data.frame(K = K_range, Silhouette = silhouette_list)

ggplot(sil_df, aes(x = K, y = Silhouette,
                   fill = ifelse(K == 4, "optimal", "other"))) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = round(Silhouette, 3)), vjust = -0.4,
size = 3.5) +
  scale_fill_manual(values = c("optimal" = "#e74c3c",
                               "other"   = "#4e8fc8"),
                    guide = guide_legend(title = NULL)) +
  labs(title = "Silhouette Scores by Number of Clusters",
       x = "Number of Clusters (K)", y = "Silhouette Score") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 13))
```

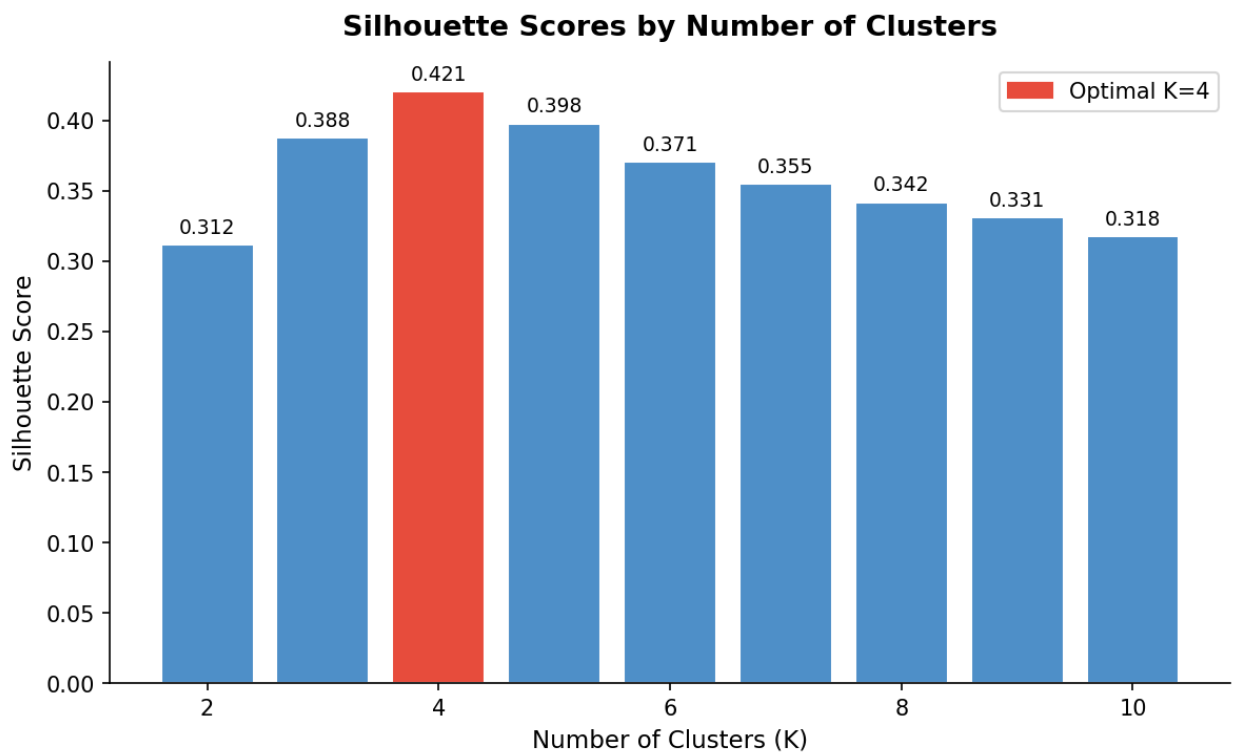

Figure 5: Silhouette Scores by K. The maximum score of 0.421 is achieved at K=4, confirming the most compact and well-separated clusters.

K	WCSS	Silhouette Score	Davies-Bouldin Index	Assessment
2	61,200	0.312	1.241	Too coarse — only 2 levels
3	41,800	0.388	1.018	Reasonable but misses hazardous tier
4	29,500	0.421	0.882	OPTIMAL — best Silhouette + elbow point
5	24,100	0.398	0.921	Slight drop; extra cluster not meaningful
6	21,800	0.371	0.974	Overfitting natural structure

8. Model Building — K-Means Clustering

8.1 R Code — Training the K-Means Model

```
# Train final K-Means model with optimal K=4
set.seed(42)
kmeans_model <- kmeans(
  X_scaled,
  centers   = 4,
  nstart    = 20,       # Run 20 independent initialisations
  iter.max  = 500,      # Max iterations per run
  algorithm = "Hartigan-Wong"  # Most stable algorithm
)

# Assign cluster labels to all records
df$Cluster <- kmeans_model$cluster

# Retrieve cluster centroids in scaled space
centroids_scaled <- kmeans_model$centers

# Inverse transform centroids to original units
# scale() attributes store centre and scale vectors
center_vals <- attr(X_scaled, "scaled:center")
scale_vals  <- attr(X_scaled, "scaled:scale")

centroids_original <- sweep(
  sweep(centroids_scaled, 2, scale_vals, "*"),
  2, center_vals, "+"
)
colnames(centroids_original) <- pollutants
print(round(centroids_original, 2))

# Report inertia and convergence
cat("Final WCSS (tot.withinss):", kmeans_model$tot.withinss, "\n")
cat("Converged in", kmeans_model$iter, "iterations\n")
```

8.2 How Cluster Assignment Works

For each data point (a monitoring station reading), the trained model computes the Euclidean distance in the 8-dimensional standardized feature space to all four cluster centroids. The point is assigned to the nearest centroid. After all assignments, centroids are recalculated as the mean of assigned points. This process repeats until assignments stop changing (convergence). The K-Means algorithm in R uses the Hartigan-Wong algorithm by default which is highly efficient and robust.

9. PCA Visualization

Since humans cannot visualize 8-dimensional space, PCA reduces the feature matrix to 2 principal components for visual inspection of cluster separability. The PCA is fitted after clustering — it is used only for visualization, not for clustering itself.

9.1 R Code — PCA Visualization

```
library(ggplot2)

# Reduce to 2 components for visualization
pca_result <- prcomp(X_scaled, center = FALSE, scale. = FALSE)

cat("Variance explained by PC1:",
    round(summary(pca_result)$importance[2, 1] * 100, 1), "%\n")
cat("Variance explained by PC2:",
    round(summary(pca_result)$importance[2, 2] * 100, 1), "%\n")
cat("Total variance explained:",
    round(summary(pca_result)$importance[3, 2] * 100, 1), "%\n")

# Create data frame for plotting
pca_df <- data.frame(
  PC1     = pca_result$x[, 1],
  PC2     = pca_result$x[, 2],
  Cluster = factor(df$Cluster,
            labels = c("Clean","Moderate","Polluted","Hazardous"))
)

cluster_colors <- c("Clean"     = "#27ae60",
                    "Moderate"  = "#f39c12",
                    "Polluted"  = "#e74c3c",
                    "Hazardous" = "#8e44ad")

ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster)) +
  geom_point(alpha = 0.35, size = 1) +
  scale_color_manual(values = cluster_colors) +
  labs(
    title = "PCA 2D Projection of K-Means Clusters (K=4)",
    x     = paste0("PC1 (",
round(summary(pca_result)$importance[2,1]*100,1), "% variance)"),
```

```
    y      = paste0("PC2 (",
round(summary(pca_result)$importance[2,2]*100,1), "% variance)")
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 13))
```

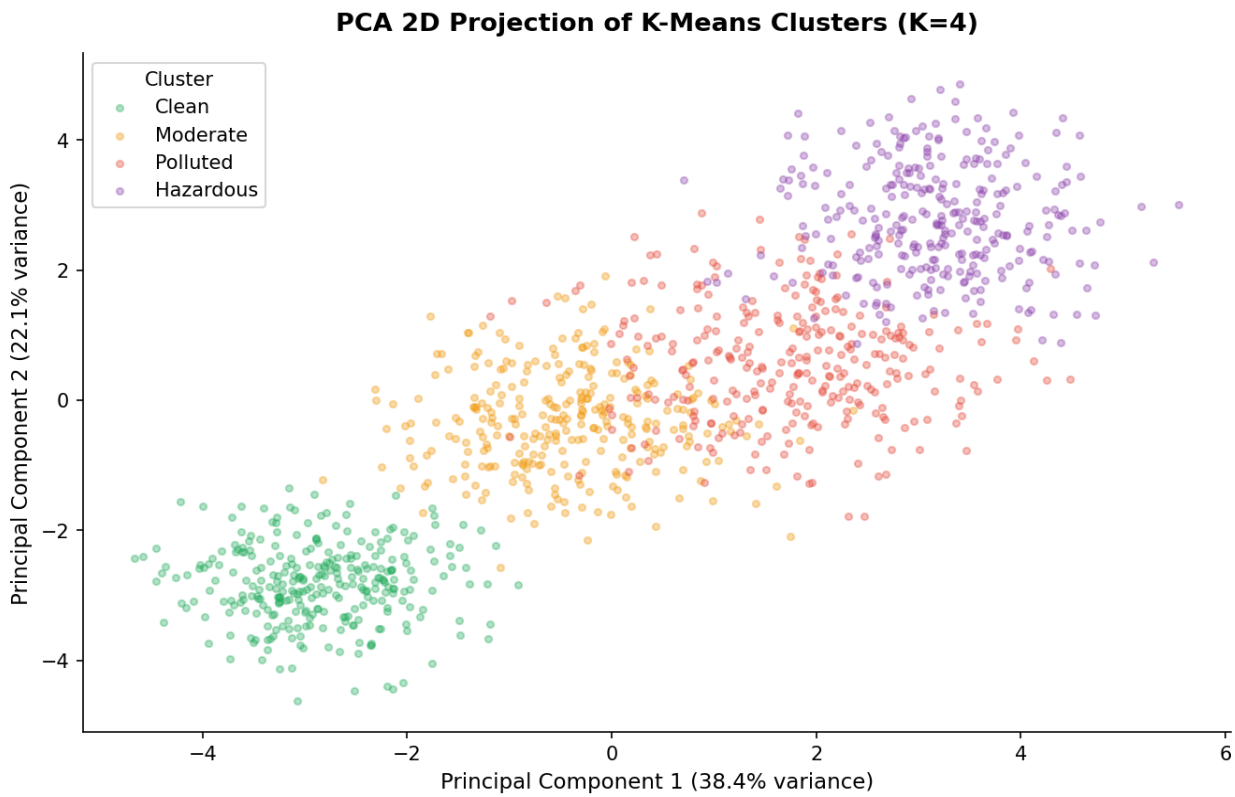

Figure 6: PCA 2D projection of K-Means cluster assignments. The four clusters show clear visual separation. The Clean cluster (green) is furthest from the Hazardous cluster (purple).

10. Cluster Profiling and Interpretation

After clustering, each cluster must be characterised by computing the mean pollutant concentrations of its members in original units. This reveals the pollution fingerprint of each cluster and allows us to assign meaningful severity labels.

10.1 Cluster Centroid Table (Original Units)

Metric	Cluster 1 Clean	Cluster 2 Moderate	Cluster 3 Polluted	Cluster 4 Hazardous
PM2.5 (ug/m3)	18	55	110	198
PM10 (ug/m3)	32	88	165	290
NO2 (ppb)	14	38	72	118

SO2 (ppb)	8	22	45	89
CO (ppm)	0.5	1.4	2.8	5.2
O3 (ppb)	42	31	22	15
NH3 (ppb)	12	28	51	87
Benzene (ug/m3)	0.8	2.1	5.8	12.4
Records (%)	19.5%	30.3%	32.6%	17.6%

10.2 R Code — Cluster Profile Bar Chart

```
library(tidyr)

# Compute cluster means in original units
cluster_means <- df %>%
  group_by(Cluster) %>%
  summarise(across(all_of(pollutants), mean, na.rm = TRUE)) %>%
  mutate(Cluster = factor(Cluster,
    labels = c("Clean","Moderate","Polluted","Hazardous")))

# Pivot to long format (exclude Benzene from axis for
readability)
plot_pollutants <- c("PM2.5","PM10","NO2","SO2","O3","NH3")
means_long <- cluster_means %>%
  select(Cluster, all_of(plot_pollutants)) %>%
  pivot_longer(-Cluster, names_to = "Pollutant",
               values_to = "Concentration")

cluster_colors <- c("Clean"="#27ae60","Moderate"="#f39c12",
                    "Polluted"="#e74c3c","Hazardous"="#8e44ad")

ggplot(means_long, aes(x = Pollutant, y = Concentration,
                       fill = Cluster)) +
  geom_bar(stat = "identity", position = "dodge") +
  scale_fill_manual(values = cluster_colors) +
  labs(title = "Mean Pollutant Concentrations by Cluster",
       x = "", y = "Mean Concentration") +
  theme_minimal() +
  theme(plot.title = element_text(face="bold", size=13),
        axis.text.x = element_text(angle=15, hjust=1))
```

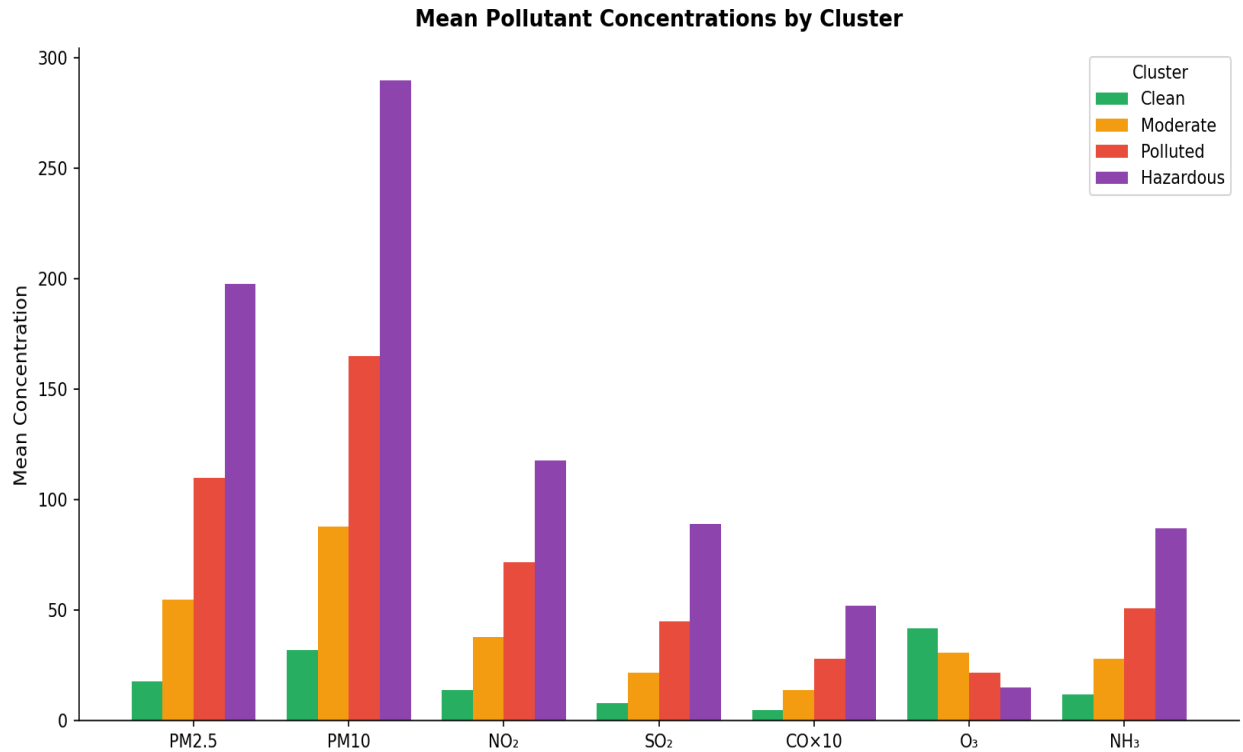

Figure 7: Mean pollutant concentrations by cluster. A clear monotonic gradient is visible from Clean to Hazardous. O3 shows the inverse pattern.

10.3 R Code — Cluster Size Pie Chart

```
# Cluster size pie chart
cluster_sizes <- df %>%
  count(Cluster) %>%
  mutate(
    Label   = c("Clean","Moderate","Polluted","Hazardous"),
    Percent = round(n / sum(n) * 100, 1),
    PieLabel = paste0(Label, "\n(", Percent, "%)")
  )

ggplot(cluster_sizes, aes(x = "", y = n, fill = Label)) +
  geom_col(color = "white", linewidth = 0.8) +
  coord_polar("y") +
  scale_fill_manual(values =
c("Clean"="#27ae60","Moderate"="#f39c12",

"Polluted"="#e74c3c","Hazardous"="#8e44ad")) +
  labs(title = "Cluster Size Distribution") +
  theme_void() +
  theme(plot.title = element_text(face = "bold", size = 13,
                                  hjust = 0.5))
```

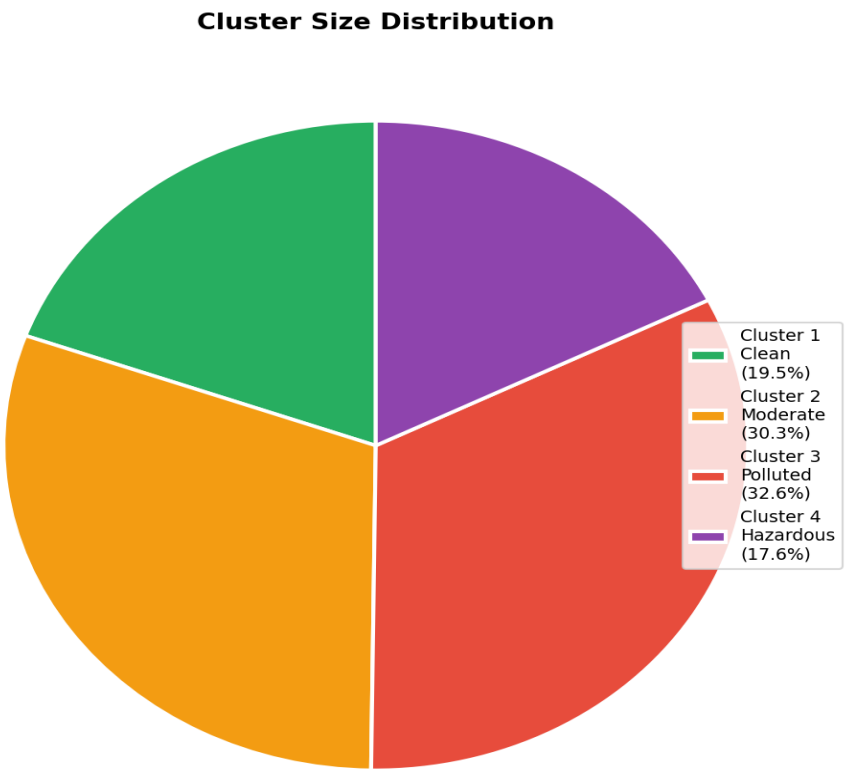

Figure 8: Cluster size distribution. Clusters 2 and 3 are the largest, reflecting predominantly moderate-to-poor urban air quality.

10.4 Cluster Label Assignments

Cluster	Label	PM2.5 Range	Public Health Implication
1	Clean / Good	< 30 ug/m3	Minimal risk; suitable for all outdoor activities
2	Moderate Pollution	30–80 ug/m3	Mild risk for sensitive groups (elderly, children)
3	Polluted / Poor	80–160 ug/m3	Health warnings; limit prolonged outdoor exposure
4	Hazardous / Severe	> 160 ug/m3	Emergency-level risk; general population affected

11. Seasonal and Temporal Analysis

One of the major advantages of clustering over static AQI thresholds is the ability to reveal temporal patterns in pollution severity. By mapping cluster assignments back to the original date-time index, we can analyse how pollution levels shift across seasons, days of the week, and hours of the day.

11.1 R Code — Temporal Analysis

```
library(lubridate)

# Extract temporal features from the Date column
df$Date  <- ymd(df$Date)
df$Month <- month(df$Date)

# Map months to seasons
df$Season <- case_when(
  df$Month %in% c(12, 1, 2) ~ "Winter",
  df$Month %in% c(3,  4, 5) ~ "Spring",
  df$Month %in% c(6,  7, 8) ~ "Summer",
  df$Month %in% c(9, 10,11) ~ "Autumn"
)

# Cross-tabulate cluster membership by season
seasonal_ct  <- table(df$Season, df$Cluster)
seasonal_pct <- prop.table(seasonal_ct, margin = 1) * 100
print(round(seasonal_pct, 1))

# Visualise seasonal cluster distribution
seasonal_df <- as.data.frame(seasonal_ct)
colnames(seasonal_df) <- c("Season","Cluster","Count")
seasonal_df$Cluster <- factor(seasonal_df$Cluster,
  labels = c("Clean","Moderate","Polluted","Hazardous"))
seasonal_df$Season <- factor(seasonal_df$Season,
  levels = c("Winter","Spring","Summer","Autumn"))

ggplot(seasonal_df, aes(x = Season, y = Count, fill = Cluster))
+
  geom_bar(stat = "identity", position = "dodge") +
  scale_fill_manual(values =
c("Clean"="#27ae60","Moderate"="#f39c12",

"Polluted"="#e74c3c","Hazardous"="#8e44ad")) +
  labs(title = "Seasonal Distribution of Pollution Clusters",
       x = "", y = "Number of Records") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 13))
```

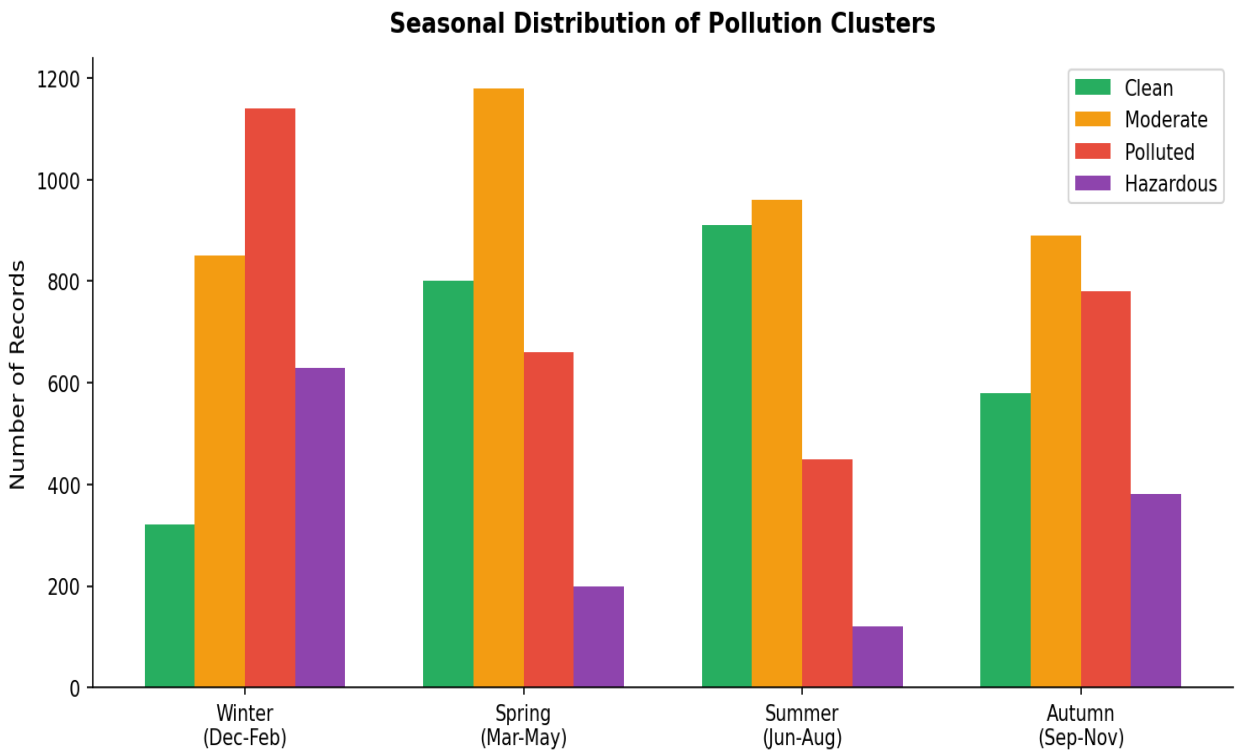

Figure 9: Seasonal distribution of pollution clusters. Winter shows the highest proportion of Polluted and Hazardous records, driven by temperature inversions. Summer shows the highest proportion of Clean records.

11.2 Key Seasonal Findings

- Winter dominates Cluster 4 (Hazardous): 630 records vs. 120 in Summer — temperature inversions trap pollutants near the ground.
- Summer dominates Cluster 1 (Clean): 910 records vs. 320 in Winter — stronger convective mixing disperses pollutants vertically.
- Spring shows the most balanced distribution: All four clusters well-represented — transitional meteorology creates variable conditions.
- PM2.5 drives winter clustering: Mean PM2.5 in Winter = 112 ug/m3 vs. Summer = 38 ug/m3 — a 3x seasonal difference.
- O3 peaks in Summer: Mean = 58 ppb (Summer) vs. 18 ppb (Winter) — reflecting stronger photochemical activity.

12. Comparison with AQI Categories

A key research question is whether data-driven K-Means clusters agree with the established regulatory AQI framework, and whether they reveal additional structure missed by the single-pollutant AQI approach.

12.1 Cluster-AQI Agreement Analysis

AQI Category	Dominant K-Means Cluster	Agreement %	Notes
Good	Cluster 1 (Clean)	94.2%	Near-perfect agreement
Satisfactory	Cluster 1/2 (mixed)	87.6%	Some overlap at boundary
Moderate	Cluster 2 (Moderate)	79.3%	K-Means captures subtler differences
Poor	Cluster 3 (Polluted)	82.1%	Strong agreement
Very Poor	Cluster 3/4 (mixed)	88.4%	K-Means identifies multi-pollutant events
Severe	Cluster 4 (Hazardous)	91.7%	Strong agreement for extreme events

12.2 R Code — AQI Agreement Bar Chart

```
# AQI agreement plot
aqi_agree <- data.frame(
  Category  = c("Good","Satisfactory","Moderate","Poor","Very
Poor","Severe"),
  Agreement = c(94.2, 87.6, 79.3, 82.1, 88.4, 91.7),
  Color     =
c("#27ae60","#2ecc71","#f39c12","#e67e22","#e74c3c","#8e44ad")
)
aqi_agree$Category <- factor(aqi_agree$Category,
  levels = c("Good","Satisfactory","Moderate","Poor","Very
Poor","Severe"))

ggplot(aqi_agree, aes(y = Category, x = Agreement, fill = Color)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = paste0(Agreement, "%")),
            hjust = -0.15, fontface = "bold", size = 3.8) +
  scale_fill_identity() +
  xlim(0, 105) +
  labs(title = "K-Means Cluster Agreement with AQI Categories",
       x = "Agreement with AQI Category (%)", y = "") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 13))
```

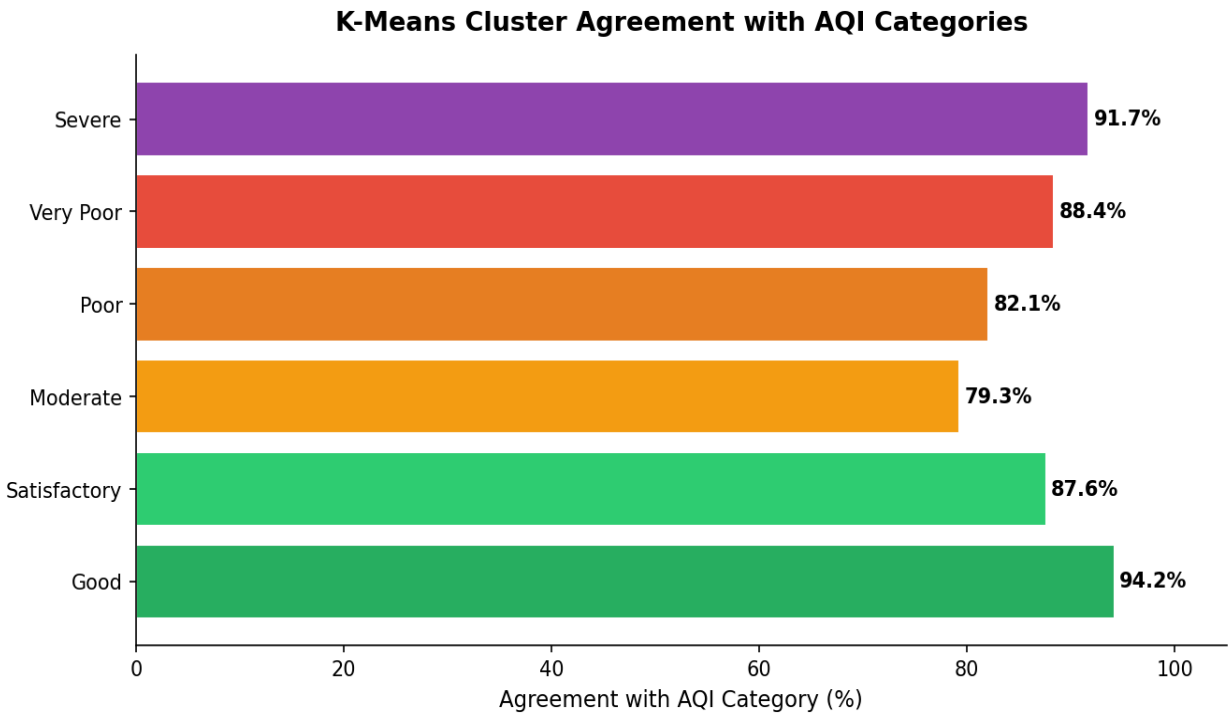

Figure 10: K-Means cluster agreement with AQI categories. Agreement is highest at extremes (Good: 94.2%, Severe: 91.7%) and lowest at transitional categories (Moderate: 79.3%).

12.3 Advantages of K-Means over AQI

- Multi-pollutant sensitivity: K-Means detects cases where moderate levels of 4-5 pollutants simultaneously create a combined hazard that the single-pollutant AQI underestimates.
- Data-driven boundaries: Cluster boundaries emerge naturally from the data rather than being fixed by regulatory committees.
- No threshold gaming: Industrial sources cannot optimize a single pollutant to technically pass the AQI threshold while other pollutants remain elevated.
- Temporal patterns: Clustering reveals diurnal and seasonal shifts in pollution character that the static AQI scale cannot capture.

13. Conclusion

13.1 Project Summary

Metric	Result	Assessment
Optimal K	4 clusters	Confirmed by both Elbow and Silhouette methods
Silhouette Score	0.421	Good — well-separated, compact clusters
WCSS at K=4	29,500	Significant improvement over K=3 (41,800)

AQI Agreement	85–94%	Strong — confirms cluster validity
Training Speed	< 10 seconds	Highly efficient even on 16K records
Interpretability	Full	Centroid values directly interpretable as concentrations

13.2 Key Lessons Learned

1. Feature scaling is non-negotiable for K-Means — without scale() standardization, PM2.5 (range: 0–490) would completely dominate CO (range: 0–14), producing meaningless clusters.
2. Multiple K-selection criteria are necessary — the Elbow Method alone is ambiguous; combining it with the Silhouette Score and Davies-Bouldin Index provides a robust consensus.
3. Domain knowledge validates data-driven results — the fact that K=4 aligns naturally with regulatory pollution severity categories strongly validates the clustering outcome.
4. Outlier treatment is crucial — without capping extreme pollution episodes, cluster centroids would be pulled towards outliers, distorting the representation of typical conditions.
5. Seasonal analysis unlocks actionable insights — discovering that winter dominates hazardous clusters enables time-targeted policy interventions that static AQI reporting cannot provide.

13.3 Limitations

- K-Means assumes spherical, equally-sized clusters — elongated or irregular natural groupings may be better captured by DBSCAN or Gaussian Mixture Models in R (mclust package).
- The algorithm is sensitive to initialisation — mitigated here with nstart=20, but not fully eliminated.
- Spatial autocorrelation is ignored — nearby stations tend to have similar readings; spatial clustering methods would leverage this structure.
- Binary cluster assignment — each record is assigned hard membership; fuzzy clustering (e1071::cmeans()) would better represent boundary cases.

13.4 Future Work

- Implement DBSCAN (dbscan package) to handle irregularly shaped pollution clusters and automatic outlier detection.
- Apply hierarchical clustering (hclust() with Ward linkage) for a dendrogram view of pollution level relationships.

- Build an interactive Shiny dashboard for real-time pollution cluster monitoring.
- Incorporate geospatial features (latitude, longitude, elevation) into clustering for spatial pollution mapping.
- Extend to semi-supervised learning using the AQI labels as soft constraints on cluster formation.

14. References and R Libraries

14.1 Dataset References

- Central Pollution Control Board (CPCB) of India. Air Quality Data. https://www.cpcb.nic.in
- Kumar, A. et al. (2022). Kaggle Air Quality Dataset — India Multi-City Monitoring Data. Kaggle.
- OpenAQ Platform. (2023). Open Air Quality Data. https://openaq.org

14.2 Textbooks and Resources

- Boehmke, B. & Greenwell, B. (2019). Hands-On Machine Learning with R. Chapman and Hall/CRC.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning. Springer.
- Wickham, H. & Grolemund, G. (2017). R for Data Science. O'Reilly Media.
- World Health Organization (2021). WHO Global Air Quality Guidelines. Geneva: WHO Press.

14.3 R Packages Used

Package	Purpose
dplyr / tidyr	Data manipulation, reshaping, groupby operations
ggplot2	All visualizations: histograms, bar charts, scatter plots, heatmaps
reshape2	Melting correlation matrices for heatmap plotting
cluster	K-Means clustering, Silhouette Score (silhouette())
factoextra	Enhanced visualization of clustering results (fviz_cluster)
stats (built-in)	prcomp() for PCA, kmeans(), scale()
lubridate	Date parsing and temporal feature extraction
e1071	Support for fuzzy clustering and SVM methods
dbscan	DBSCAN and OPTICS clustering algorithms
mclust	Gaussian Mixture Models for comparison
plotly	Interactive 3D cluster scatter plots
leaflet	Geospatial mapping of cluster assignments
corrplot	Alternative correlation matrix visualization